

Experiment – 6

Date: 10/10/2026

Aim of the Experiment:

To interface and control a servo motor with Arduino for light-level indication.

Objective:

- 1) Familiarization of Servomotor, Comparison with other motors, and Interfacing of servo motor with Arduino
- 2) Controlling Servo Motor with and without potentiometer.
- 3) Implementation of a visual Dial which indicates the environmental light conditions.

Components/Equipment Required:

Sl No.	Name of the Component/Equipment	Specification	Quantity
1	Arduino UNO R3	16MHz	1
2	Arduino UNO cable	USB Type A to B	1
3	Resistors (carbon type)	¼ watt (220Ω)	2
		¼ watt (4.7kΩ)	1
4	LDR/photoresistor		1
5	Potentiometer	10k	1
6	Buzzer	passive	1
7	Breadboard	840 Tie points	1
8	Jumper Wire	-----	As per requirement
9	Servo Motor	SG90	2

Theory:

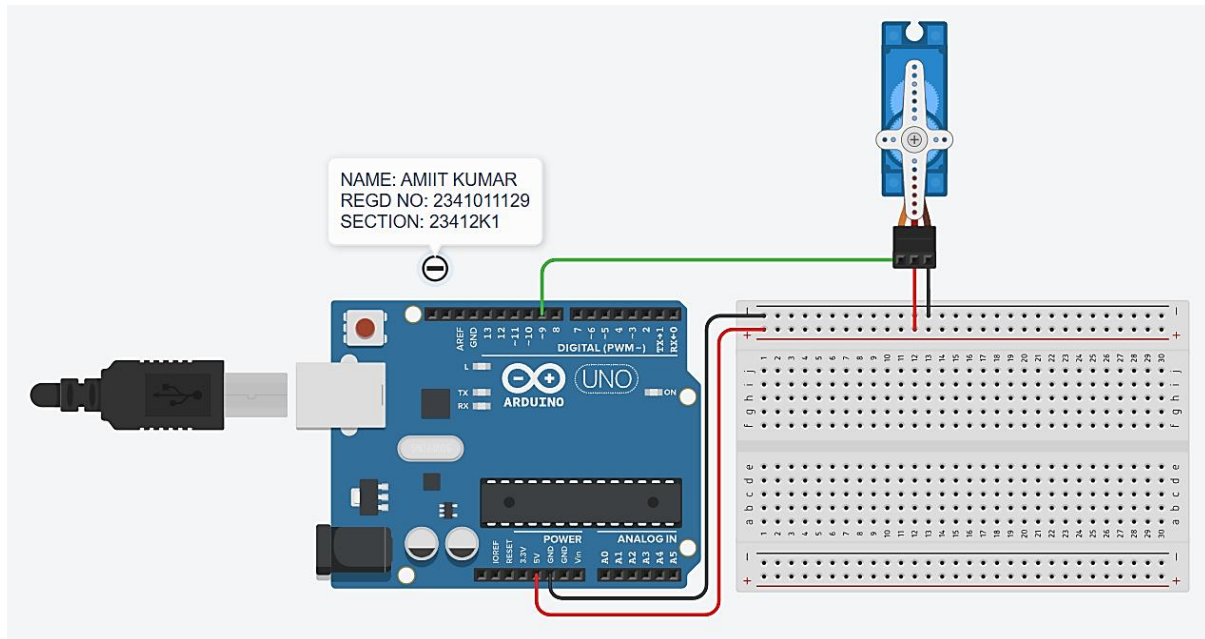
- Servo Motor: A servo motor provides precise angular position control using a motor, gears, feedback system, and control circuit.
- SG90 Servo: The SG90 typically provides approximately 0°–180° angular movement and is controlled by PWM signals.

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- Arduino Control: Arduino controls the servo using the Servo library by specifying the required angle.
- LDR: An LDR is a light-sensitive resistor whose resistance changes according to the intensity of incident light.
- LDR Response: LDR resistance generally decreases with increasing light and increases in darkness.
- Potentiometer: A potentiometer provides a variable analog voltage that can be used to control servo position.
- Visual Dial: The experiment uses the servo as a pointer to visually indicate environmental light conditions.

Objective 1

Circuit Diagram (Tinker CAD):



Code/Program:

```
#include <Servo.h>

int servoPin = 9;
int delayTime = 4000;

Servo myServo;

void setup() {
  myServo.attach(servoPin);
  Serial.begin(9600);
}

void loop() {
  myServo.write(0);
  delay(delayTime);

  myServo.write(10);
  delay(delayTime);

  myServo.write(45);
  delay(delayTime);

  myServo.write(60);
  delay(delayTime);

  myServo.write(120);
  delay(delayTime);

  myServo.write(135);
  delay(delayTime);

  myServo.write(150);
  delay(delayTime);

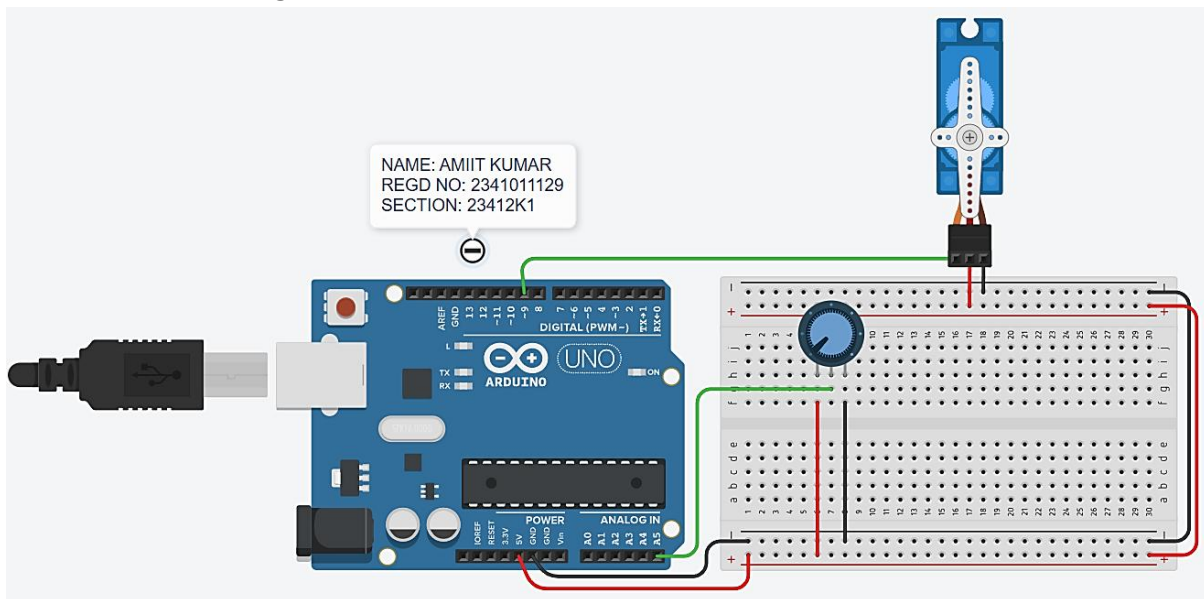
  myServo.write(180);
  delay(delayTime);
}
```

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Observation: The SG90 servo motor responded to the control signal from the Arduino and moved to the commanded angular position. Unlike a normal DC motor, the servo provided controlled angular positioning rather than continuous uncontrolled rotation.

Objective 2

Circuit/Schematic Diagram:



A) With Potentiometer

Code:

```
#include <Servo.h>

int potPin = A5;
int servoPin = 9;
Servo myServo;
int val1;
int val2;

void setup() {
  Serial.begin(9600);
  myServo.attach(servoPin);
  pinMode(potPin, INPUT);
}

void loop() {
  val1 = analogRead(potPin);
  Serial.print(val1);
  Serial.print(" ");

  val2 = map(val1, 0, 1023, 0, 180);
  myServo.write(val2);

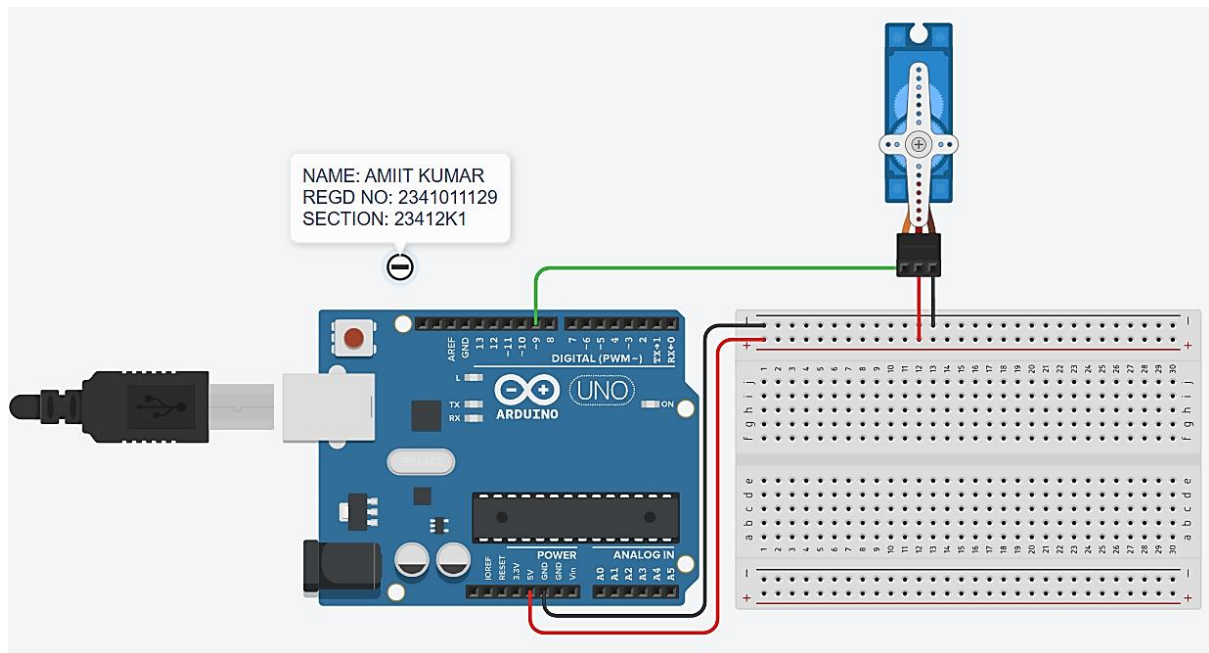
  Serial.println(val2);
  delay(500);
}
```

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Observation: When the potentiometer was rotated, its analog output changed. The Arduino converted this analog value into a corresponding servo angle, causing the servo to rotate proportionally.

Objective 2

Circuit/Schematic Diagram:



B) Without Potentiometer

Code:

```
#include <Servo.h>

Servo myServo;
int servoPos = 0;
int servoPin = 9;
int delayTime = 300;

void setup() {
  myServo.attach(servoPin);
}

void loop() {
  for (servoPos = 0; servoPos <= 180; servoPos += 30) {
    myServo.write(servoPos);
    delay(delayTime);
  }

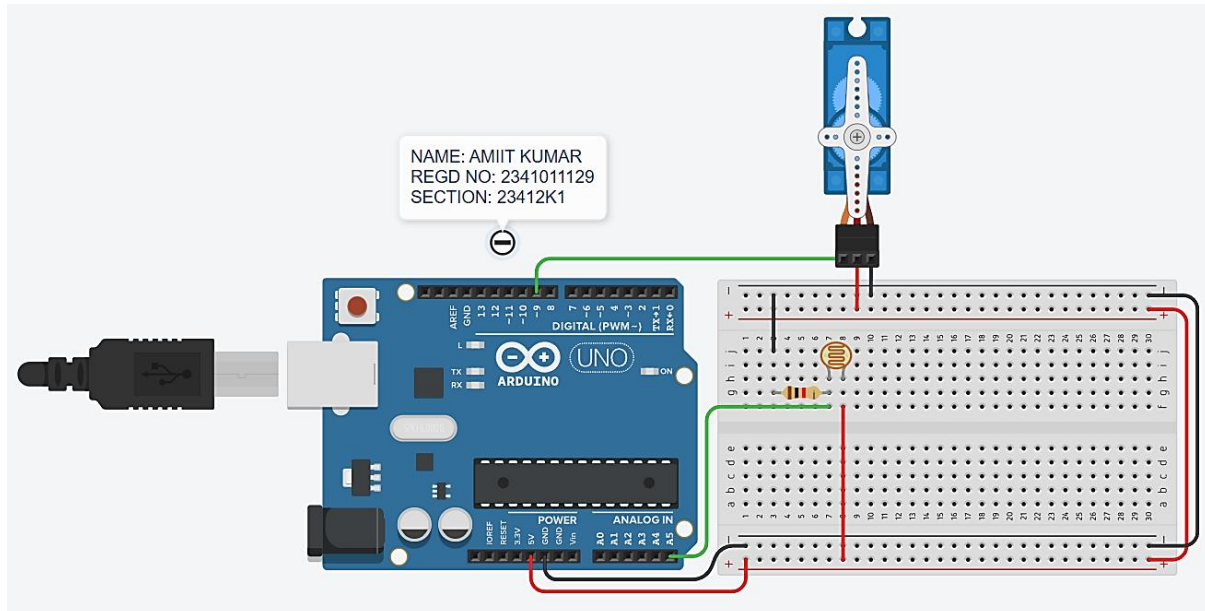
  for (servoPos = 180; servoPos >= 0; servoPos -= 30) {
    myServo.write(servoPos);
    delay(delayTime);
  }
}
```

Observation: Without a potentiometer, the servo angle was directly specified in the Arduino program. The servo moved automatically to the programmed positions.

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Objective 3

Circuit/Schematic Diagram:



Code:

```
#include <Servo.h>

int servoPin = 9;
int ldrPin = A5;
int lightVal;
int delayTime = 250;
int angleServo;
Servo myServo;

void setup() {
  Serial.begin(9600);
  pinMode(ldrPin, INPUT);
  myServo.attach(servoPin);
}

void loop() {
  lightVal = analogRead(ldrPin);
  delay(delayTime);

  angleServo = (-16./63.) * lightVal + 160. * (26./21.);
  myServo.write(angleServo);
  Serial.println(angleServo);
}
```

Observation: The servo pointer changed its angular position according to the light detected by the LDR. Thus, the servo successfully acted as a visual indicator of environmental light conditions.

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Post Experiment Questionnaire:

- 1) Which library is used for efficient control of SG90 servo motor in Arduino sketch?

Solⁿ The Servo library is used to control the SG90 servo motor.

It provides functions such as: `#include <Servo.h>` and allows the servo to be attached and positioned using functions such as `attach()` and `write()`.

- 2) Develop a mathematical equation to explain the relationship between LDR light condition and servo angle.

Solⁿ Let L be the LDR analog reading (0–1023) and θ be the servo angle (0°–180°). The linear mapping is $\theta = (L \times 180)/1023$. For $L = 512$, $\theta \approx 90^\circ$. If the voltage-divider connection makes the reading decrease as light increases, use $\theta = 180 - (L \times 180)/1023$.

- 3) Write the syntax of `map()` function in Arduino sketch. Where is it used? Explain it through one example.

Solⁿ Syntax: `map(value, fromLow, fromHigh, toLow, toHigh)`. The `map()` function converts a value from one numerical range to another. For example, `int angle = map(analogRead(A0), 0, 1023, 0, 180);` converts a sensor reading of about 512 into a servo angle of about 90°.

- 4) Can we replace the servomotor with any other motor in various engineering applications? Justify your answer.

Solⁿ A servo motor cannot always be directly replaced by another motor. A servo provides feedback-based angular position control, whereas a DC motor usually provides continuous rotation and a stepper motor moves in discrete steps. Replacement is justified only when the alternative motor and its controller meet the application's torque, accuracy, speed, and positioning requirements.

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